GENAI BASICS • LESSON 08 • MODULE 3: AI ASSISTANTS AT WORK

AI for Data Analysis

Spreadsheet → answer → chart → one-page report

Formulas, pivots and charts on request in plain words — in Excel, Google Sheets or a chat with a file. Case study: Northstar online shop sales for 2025.



WHERE ANALYSIS STARTS

The manager’s question comes first, then the file

AN E-MAIL FROM ANNA, COMMERCIAL DIRECTOR

Hi! How did sales go this year? Why did advertising pay off worse in the second half? And why so many returns in the autumn? I need one page by Friday.

> Three questions — three charts. “Find insights” won’t help here.

NORTHSTAR_2025.XLSX • 1,261 ROWS

A1 | order_date

	A	B	C	D	E	F	G	H	I
1	order_date	channel	category	region	warehouse	orders	revenue	ad_spend	returns
2	2025-01-01	Social	Home	Central	C1	10	580.00	135.10	0
3	2025-01-02	Organic	Sports	West	W1	11	670.00	0.00	0
4	2025-01-02	Social	Books	West	W1	18	400.00	103.34	1
5	2025-01-02	Social	Home	West	W1	16	890.00	222.15	1
6	2025-01-03	Organic	Electronics	South	S1	39	3,660.00	0.00	2
7	2025-01-03	Organic	Sports	North	N1	19	1,170.50	0.00	1
8	2025-01-03	Search	Electronics	Central	C1	36	3,500.00	626.28	2
9	2025-01-04	Email	Home	North	N1	24	1,210.50	0.00	1
10	2025-01-04	Organic	Home	West	W1	11	610.50	0.00	1
11	2025-01-04	Social	Books	West	W1	22	490.50	117.76	1
12	2025-01-04	Social	Electronics	North	N2	35	3,250.50	795.72	1
13	2025-01-05	Email	Electronics	North	N2	11	980.00	0.00	1
14	2025-01-05	Search	Home	Central	C1	47	2,480.50	449.10	2
15	2025-01-05	Search	Home	South	S1	68	3,440.50	632.18	3
16	2025-01-05	Social	Home	West	W1	18	1,030.50	254.28	1
17	2025-01-06	Organic	Electronics	West	W1	9	840.50	0.00	0
18	2025-01-06	Organic	Home	West	W1	17	970.50	0.00	1
19	2025-01-07	Email	Electronics	Central	C1	28	2,440.50	0.00	1
20	2025-01-07	Organic	Books	Central	C1	54	1,340.00	0.00	3
21	2025-01-08	Email	Sports	Central	C1	23	1,390.50	0.00	1
22	2025-01-08	Email	Sports	South	S1	16	920.00	0.00	1
23	2025-01-08	Email	Sports	West	W1	13	820.00	0.00	1
24	2025-01-08	Search	Home	West	W1	30	1,610.00	300.21	1
25	2025-01-09	Email	Electronics	South	S1	30	2,810.50	0.00	2
26	2025-01-09	Search	Books	South	S1	31	730.50	142.49	1
27	2025-01-09	Search	Sports	Central	C1	18	1,080.00	209.35	1

Sales | Reviews | Ad ROAS 2025

One row is the sales of one channel for one category, region and warehouse on one day. The second sheet holds 357 customer reviews.

First prompt: describe the file and suggest 5 questions

PROMPT

Describe this file: what one row is, how many rows, what period, what data problems. Suggest 5 questions a manager would ask. Don’t change anything yet.

CHATGPT ANSWER • PART 1

Sales	1 261 строка, 9 колонок	Агрегированные продажи за дату по каналу, категории, региону и складу
Reviews	357 строк, 7 колонок	Один отзыв покупателя с оценкой и комментарием

Sales: дата, канал продаж, категория товара, регион, склад, количество заказов, выручка, рекламные расходы и возвраты.

Reviews: ID отзыва, дата, регион, категория, склад, рейтинг от 1 до 5 и текст комментария.

Период обоих листов: 01.01.2025–31.12.2025.

2. Проблемы с данными

- В `Sales` отсутствуют 632 значения `ad_spend` — нужно выяснить, означают ли они нулевые расходы.
- Отсутствуют 32 значения `returns`.
- Обнаружена одна полностью дублирующаяся строка продаж.
- В `Reviews` явных пропусков и повторяющихся ID нет.
- Нет идентификатора заказа, поэтому связать конкретный отзыв с конкретной покупкой невозможно.

CHATGPT ANSWER • PART 2

3. Пять вопросов для руководителя

1. **Какие каналы продаж наиболее прибыльны?** С учётом рекламных расходов.
2. **Какие категории товаров обеспечивают основной рост выручки?**
3. **Где самые высокие показатели возвратов?** По категориям, регионам и складам.
4. **Связаны ли низкие оценки покупателей с высоким уровнем возвратов?**
5. **Какие регионы и каналы стоит масштабировать в 2026 году?**

Вывод: данные подходят для анализа продаж, эффективности рекламы, возвратов и клиентского опыта. Перед расчётом KPI потребуется уточнить значение пропусков и проверить дубликат.

Файл не изменял, бизнес-показатели не рассчитывал.

The model found the duplicate and the period and said: **“I did not change the file”**. But it took zero ad spend for missing values — that is the first question for the data owner.

We take question 1, channels including ad spend — it matches the e-mail. That is what the demo computes.

WHERE TO ANALYSE

Pick the tool by where the data lives

Inside it is always the same: the model writes a formula, code or a query, and a program does the counting. That is why you can open and check the working.



Excel

Data already lives in a workbook. AI writes **formulas and pivots** you can see in the cells.

`Claude / Copilot in Excel, or a file from a chat`



Google Sheets

The file lives in Google Drive. Gemini in the side panel builds **pivots and charts**.

`Gemini in Sheets`



Chat with a file

Upload a CSV or XLSX. The model writes **Python code**, a program runs it, and you see the code.

`ChatGPT · Claude · Gemini`



Database

Millions of rows that won’t fit in a chat. The model writes **an SQL query**, the database does the counting.

`SQL + an AI assistant`

your question
in plain words → the model writes
formula · code · SQL → a program counts
Excel · Python · database → you open and check
the cell · code · query

In today’s demos: Excel and a chat with a file. The channel database example is on slide 13.

DEMO 1 • EXCEL • 12 MINUTES

You can ask for a formula and a chart in words

PROMPT 2

Add a sheet “Ad return” to the workbook: by month, revenue, ad spend and revenue per $1 of ads for Search and Social. Use Excel formulas, not pasted numbers. Build a line chart and let me download the .xlsx.

Have Claude or Copilot in Excel? Type the same text in the side panel; nothing to download.

CHATGPT RETURNED A SHEET: NUMBERS IN THE CELLS

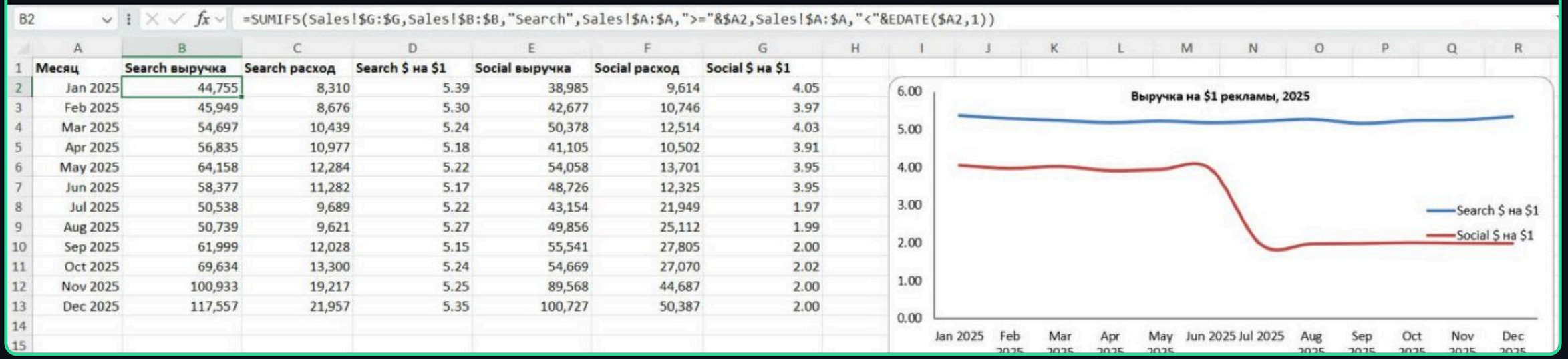


A1 fx Month

	A	B	C	D	E	F
1	Month	Search revenue	Search ad spend	Search ROAS	Social revenue	Social ad spend
2	Jan 2025	44,755.00	8,310.23	5.39x	38,985.00	9,614.32
3	Feb 2025	45,949.00	8,676.11	5.30x	42,677.00	10,746.11
4	Mar 2025	54,696.50	10,439.37	5.24x	50,378.00	12,514.00
5	Apr 2025	56,835.00	10,977.28	5.18x	41,104.50	10,501.66
6	May 2025	64,157.50	12,283.61	5.22x	54,057.50	13,700.63
7	Jun 2025	58,377.00	11,281.95	5.17x	48,726.00	12,325.24
8	Jul 2025	50,537.50	9,689.36	5.22x	43,154.00	21,948.53
9	Aug 2025	50,739.00	9,620.82	5.27x	49,856.00	25,111.57
10	Sep 2025	61,998.50	12,027.74	5.15x	55,541.00	27,804.92
11	Oct 2025	69,634.00	13,300.06	5.24x	54,668.50	27,070.16
12	Nov 2025	100,933.00	19,217.13	5.25x	89,567.50	44,686.85
13	Dec 2025	117,556.50	21,956.98	5.35x	100,727.00	50,387.44
14						
15	ROAS = revenue / ad spend; revenue is attributed to channel, not proven incremental.					

Click a cell. See a number? Write: “replace the numbers with formulas”.

EXCEL: THE CELL HOLDS A FORMULA, VISIBLE IN THE FX BAR



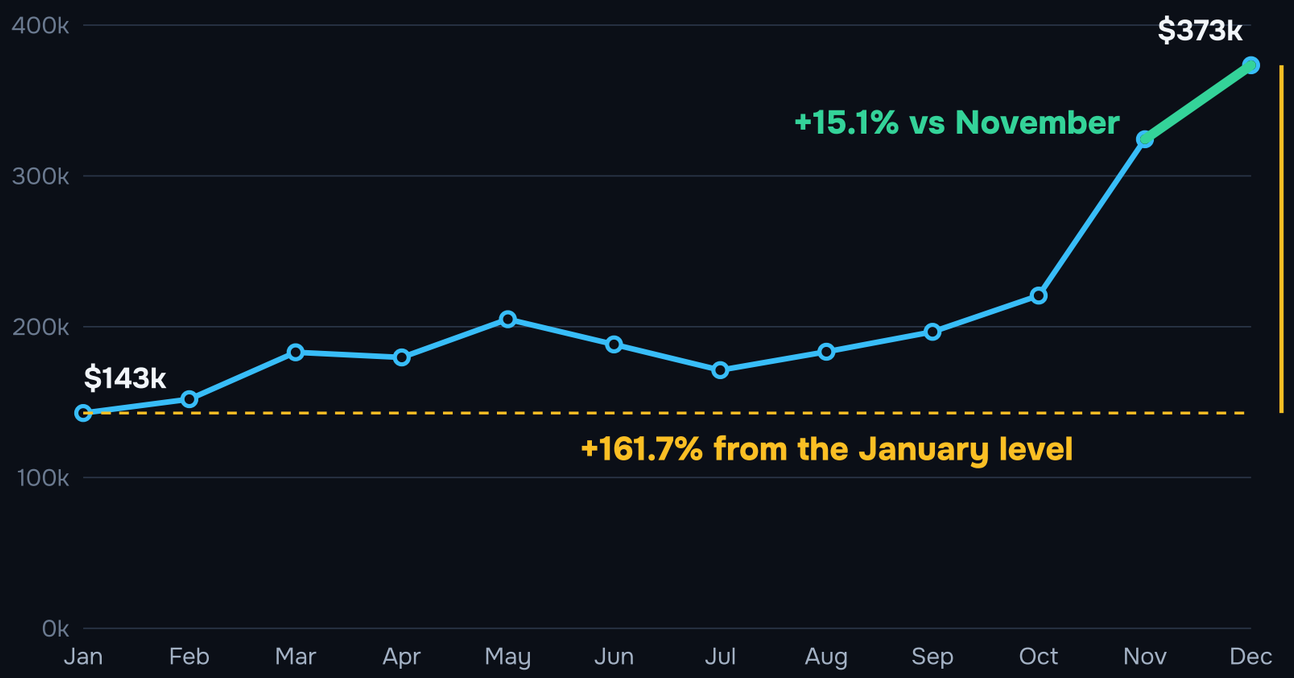
B2 fx =SUMIFS(Sales!$G:$G,Sales!$B:$B,"Search",Sales!$A:$A,">="&$A2,Sales!$A:$A,"<"&EDATE($A2,1))

	A	B	C	D	E	F	G
1	Месяц	Search выручка	Search расход	Search $ на $1	Social выручка	Social расход	Social $ на $1
2	Jan 2025	44,755	8,310	5.39	38,985	9,614	4.05
3	Feb 2025	45,949	8,676	5.30	42,677	10,746	3.97
4	Mar 2025	54,697	10,439	5.24	50,378	12,514	4.03
5	Apr 2025	56,835	10,977	5.18	41,105	10,502	3.91
6	May 2025	64,158	12,284	5.22	54,058	13,701	3.95
7	Jun 2025	58,377	11,282	5.17	48,726	12,325	3.95
8	Jul 2025	50,538	9,689	5.22	43,154	21,949	1.97
9	Aug 2025	50,739	9,621	5.27	49,856	25,112	1.99
10	Sep 2025	61,999	12,028	5.15	55,541	27,805	2.00
11	Oct 2025	69,634	13,300	5.24	54,669	27,070	2.02
12	Nov 2025	100,933	19,217	5.25	89,568	44,687	2.00
13	Dec 2025	117,557	21,957	5.35	100,727	50,387	2.00
14							
15							

WHAT WE SAW • SEASONALITY

Growth this year: +161.7% or +15.1%? It depends on the base

Northstar revenue by month, 2025



+161.7%

December vs January. January is the weakest month

+15.1%

December vs November. The peak already started in November

27.7%

of annual revenue came in November and December

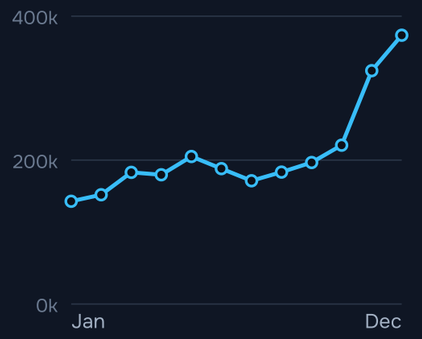
Ask: a percentage of which month, over what period?

CHARTS

Five questions, five charts — all from our file

How did it change over time?

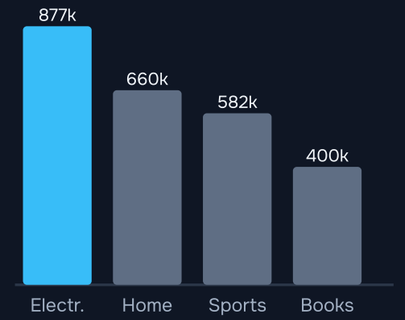
LINE



peak in Nov-Dec

Which is bigger?

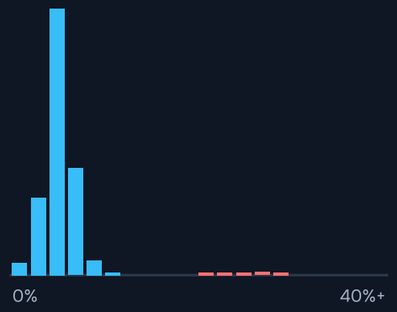
SORTED BARS



Electronics leads revenue

How are values spread?

HISTOGRAM



red tail: 22 rows with 14%+ returns

Are two numbers related?

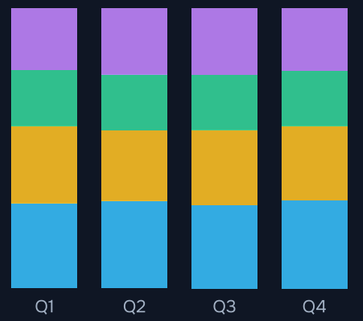
SCATTER



● Search ● Social Jan–Jun ● Social Jul–Dec: spend up, revenue not

What makes up the whole?

100% BARS



revenue shares by quarter: ● Search ● Social ● Email ● Organic

Ask like this: “Build one chart that answers …, label the axes and units.” Name the question first; the model will pick the chart type.

BREAK

5 minutes

05:00

Next: a demo in a chat with a file — returns, reviews, a slide-ready chart. Then a large database, a one-page report and homework.

DEMO 2 • CHAT WITH A FILE • 12 MINUTES

6% returns for the year. One cell shows 24.8%

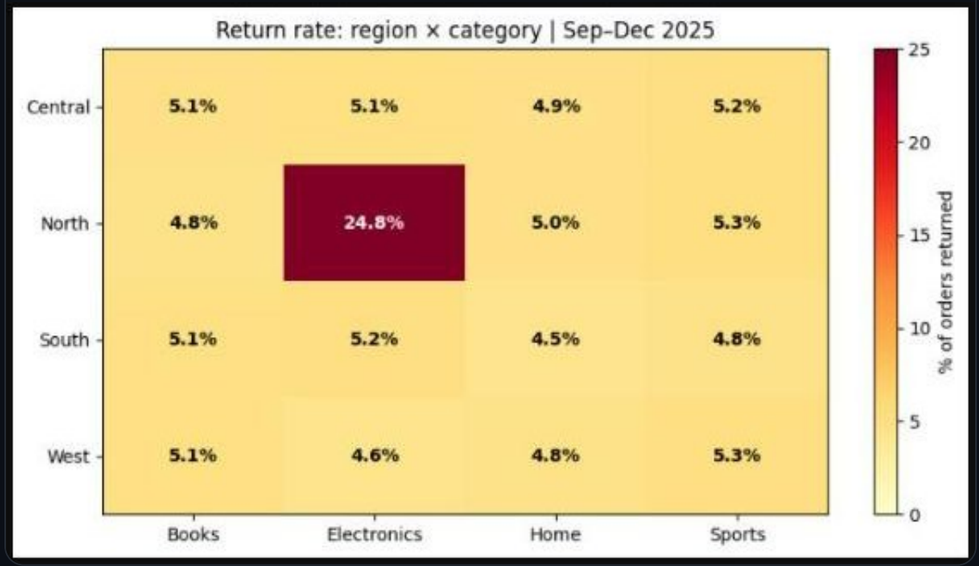
PROMPT 3

Compute the return rate by region and category for September–December and show it as a heatmap.

PROMPT 4

For the warehouse with the highest return rate, sort the reviews into themes and compare it with the other warehouses. Show 5 example reviews.

HEATMAP



REVIEW THEMES



PROMPT 5

Rework the chart for a slide. Full text and result on slide 11.

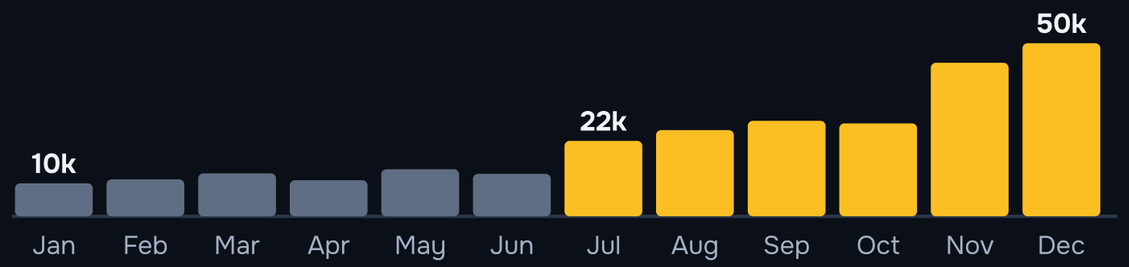
Teams chat task: which number on the heatmap would you check first?

WHAT WE FOUND

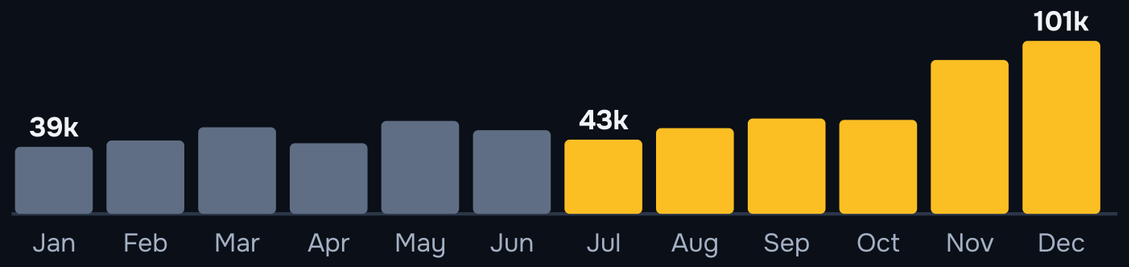
Two problems the annual total doesn’t show

1. Social: spend went up, revenue didn’t

Social ad spend by month, $



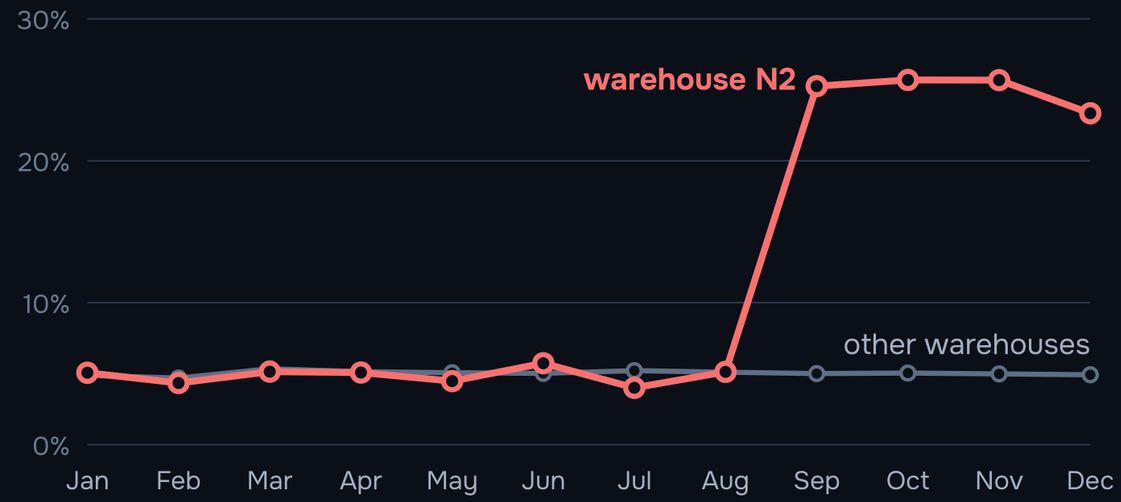
Social revenue by month, $



Second half vs first: spend **+184%**, revenue **+43%**. Revenue per $1 of ads: 4 → 2.

2. Warehouse N2: returns and damage complaints

Return rate by month, %



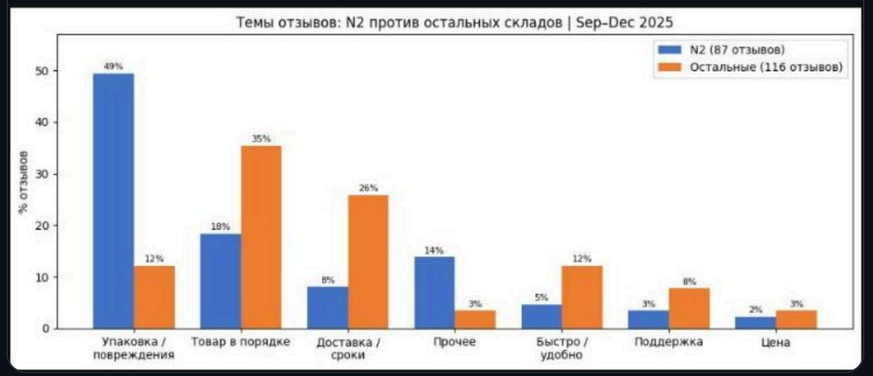
From September N2 returns are about 26%, others about 5%. Almost half of N2 reviews are about damaged parcels.

The number shows where; the text explains why. Reviews are data too.

A SLIDE-READY CHART

You edit a chart with words, like text

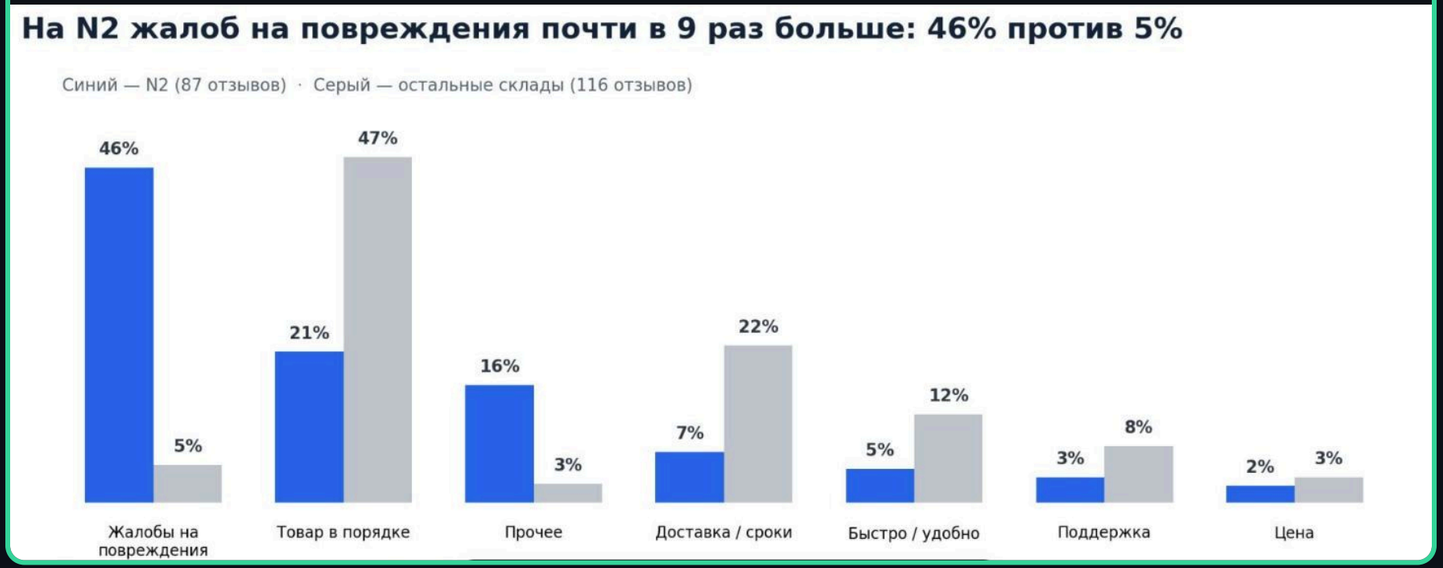
BEFORE: THE DEFAULT CHART



PROMPT

Rework this chart for a slide: the title states the finding, bars sorted, value labels on the bars, no legend and no gridlines, one accent colour for N2, everything else grey. Explain in 3 points what you changed.

AFTER: 36 SECONDS LATER



Title = finding. “Almost 9× more” instead of “Review themes”.

One colour. N2 blue, the rest grey — the eye finds the point at once.

Values on the bars, no legend or grid.

CHECK • 60 SECONDS

Three versions of a chart, three different numbers. Compare after every redo

Share of N2 reviews about damage, September–December

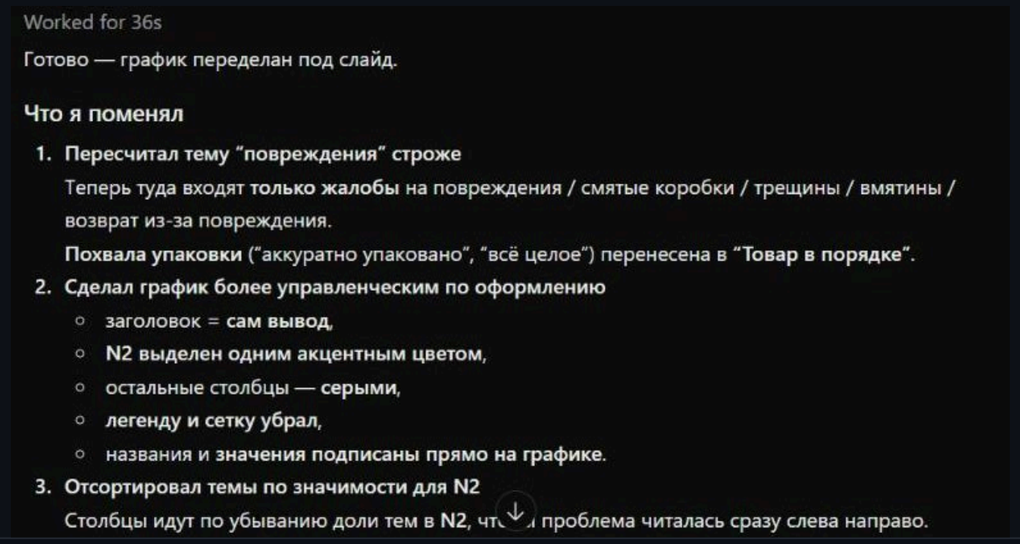
49 % first chart

46 % after restyling

44 % in the Word report

61 % by the labels the file was built from

THE MODEL EXPLAINS WHAT IT CHANGED



Worked for 36s

Готово — график переделан под слайд.

Что я поменял

1. **Пересчитал тему “повреждения” строже**
   Теперь туда входят **только жалобы** на повреждения / смятые коробки / трещины / вмятины / возврат из-за повреждения.
   Похвала упаковки (“аккуратно упаковано”, “всё целое“) перенесена в **“Товар в порядке“**.
2. **Сделал график более управленческим по оформлению**
   - заголовок = **сам вывод**,
   - **N2 выделен одним акцентным цветом**,
   - остальные столбцы — **серыми**,
   - **легенду и сетку убрал**,
   - **названия и значения подписаны прямо на графике**.
3. **Отсортировал темы по значимости для N2**
   Столбцы идут по убыванию доли тем в N2, чт… проблема читалась сразу слева направо.

1 Rows before and after

1,261 in the file → 1,260 after the duplicate is removed. How many were dropped, and why?

2 One point: against the source and the previous version

Find one number from the chart in the file and in the previous answer. Does it match?

3 Percent of what

From which month, over what period? +161% and +15% are both right.

Check review themes by reading: open 10 reviews and count yourself.

A LARGE DATABASE • COOKING YOUTUBE CHANNELS

The database updates daily: don’t export a file, ask the database

26,544
channels: name, subscribers, country

416,809
videos: length, date, channel

668,665
daily view measurements

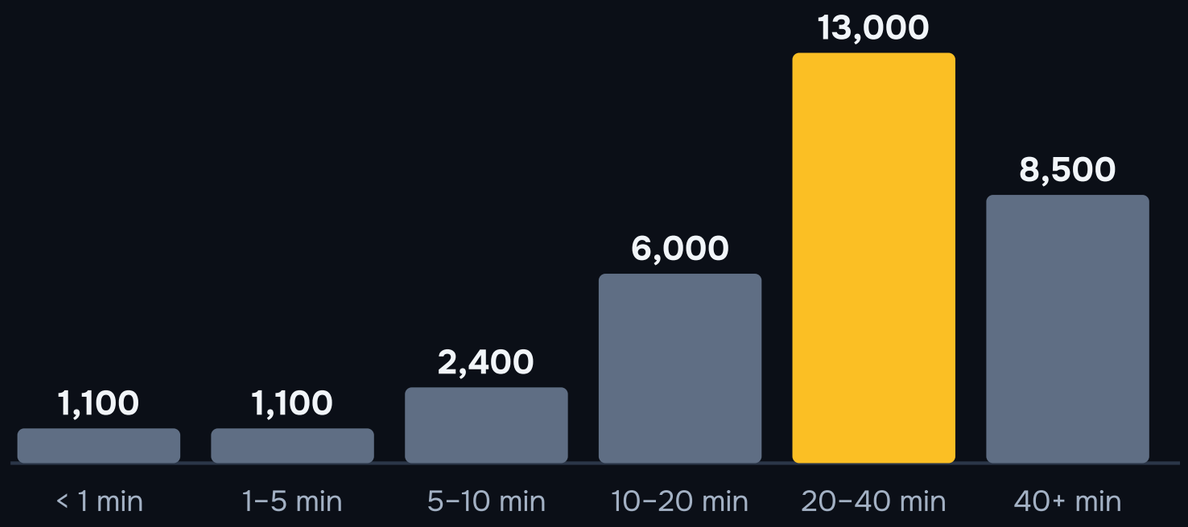
For the lesson, an anonymised sample: **1,943** channels, **46,132** videos. No e-mails, links or ids.

PROMPT

```
The database has a videos table: length and views of every
video. Write an SQL query: which video length gets the
most views? Explain every line.
```

```sql
-- Which video length gets the most views?
SELECT duration_bucket,            -- length
       COUNT(*)      AS videos,    -- how many videos
       MEDIAN(views) AS typical    -- typical views
FROM videos
GROUP BY duration_bucket;
```

Typical views per video by length



20–40 minutes gets 12× more than short videos? Compared with ordinary videos **of the same channel**, they are only 3% above usual. Big channels simply make long videos more often.

WHERE DATA COMES FROM

Look for an Export button first, then an API. Scraping is the last option

1. Export

An “Export to Excel / CSV” button exists almost everywhere: CRM, accounting, ad accounts, YouTube Studio, Google Analytics.

`fast · legal · no code`

2. Official API

```
{ "title": "Kitchen Basics",
  "subscriberCount": "184000",
  "videoCount": "412" }
```

A program asks the service and gets an answer like this. AI writes a script that turns it into table rows.

`this is how the slide 13 channel database was collected`

3. Scraping

A program reads website pages like a person would. Before you do it, check:

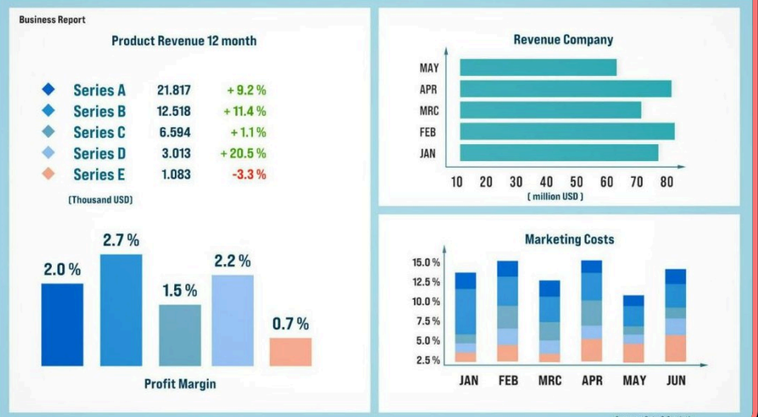
1. the site’s terms allow it
2. no personal data
3. you don’t overload the site

Ask the AI: “What official export or API does system X have to download a year of sales?”

THE RESULT OF THE ANALYSIS

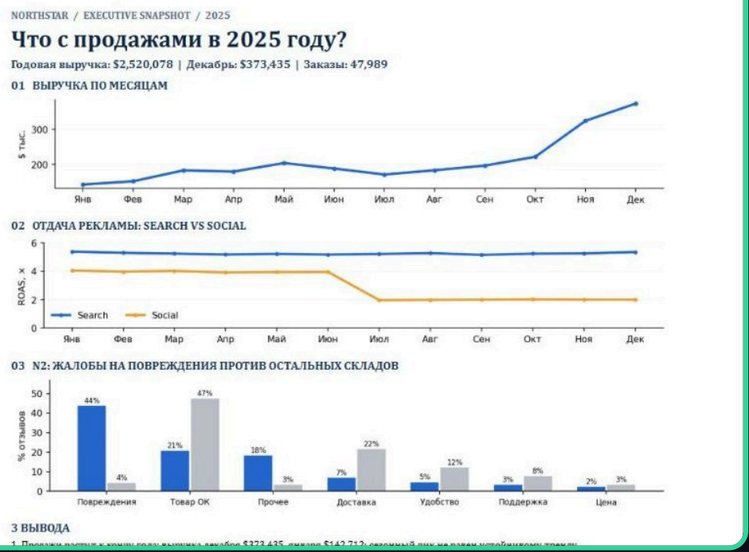
Send the manager the report file. The chat preview can lie

✗ PREVIEW IN THE CHAT



A placeholder picture: “Series A”, invented numbers. None of our analysis is here.

✓ THE DOWNLOADED .DOCX



The real file: the question, three charts, three findings, “what the data does not prove”.

PROMPT 6

```
Build a one-page report for a manager in Word: the question, 3 charts (revenue by month, ad return, N2 complaints), 3 findings and one point on “what this data does not prove”. And give me a short Python script that builds the same 3 charts from next month’s file.
```

RECIPE FOR NEXT MONTH

Bash

```
pip install pandas openpyxl matplotlib
python rebuild_northstar_charts.py next_month.xlsx
```

Результат появится в папке `charts`:

- `01_revenue.png` — выручка по месяцам.
- `02_roas.png` — отдача Search и Social.
- `03_reviews_N2.png` — темы отзывов N2 и остальных складов.

AFTER THE FINDING

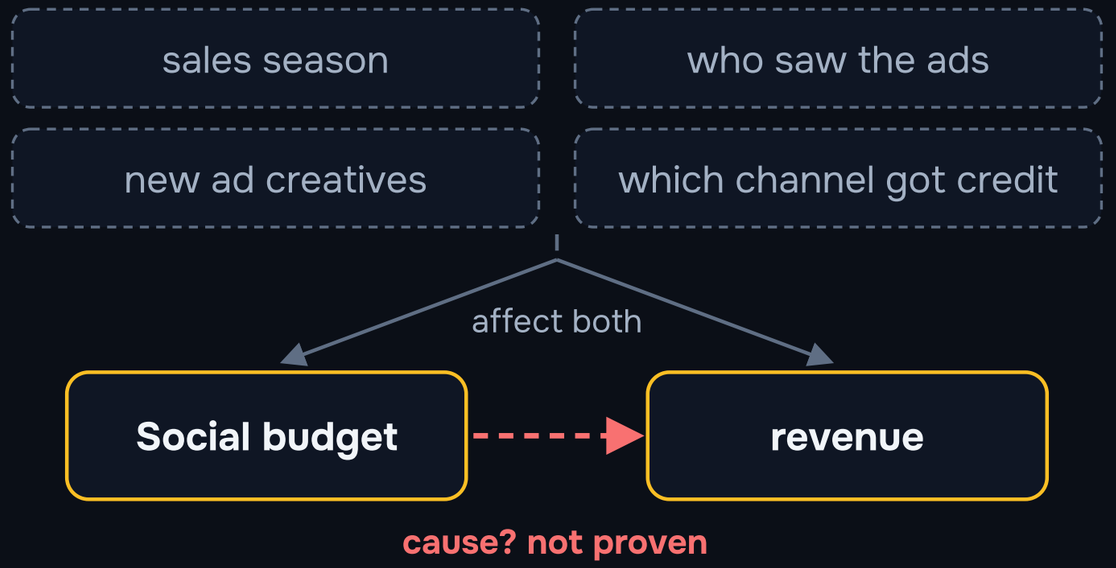
The table says what happened. Only a test shows what a budget cut will do

The temptation after the demo: **cut Social ads in half?**

The data shows: since July Social spends twice as much per dollar of revenue.

The data doesn’t show: how much revenue disappears if ads are cut.

How to find out: cut the budget in one region for 2–4 weeks and compare with the others.



DATA AND PRIVACY

What you may upload to a chat

- **OK: practice, public, anonymised data**
  like Northstar; any service
- **Careful: internal company data**
  only in the work account (ChatGPT Business/Enterprise, Claude Team/Enterprise, Google Workspace) — data is not used for training there
- **Not without approval: customers, staff, finance, health**
  personal data — only in an approved environment with the data owner's consent

Before uploading, **delete unneeded columns**: names, phones, e-mails. Sales analysis doesn't need them.

In personal accounts, training on your data **depends on settings**: ChatGPT — "Improve the model for everyone", Gemini — "Keep Activity", Claude — only if you allowed it.

PRACTICE AND HOMEWORK

A pack you can send to a manager

File `northstar_2025.xlsx`. Answer Anna's e-mail: **how did sales go this year, and what should we check next?**

```
analysis_pack/
  analysis.xlsx    # AI formula + pivot + chart
  charts/          # 3 charts, one as before / after
  report.docx      # 1 page: question, 3 findings
  recipe.md        # prompt + script for January
  checks.md        # 60 seconds + "does not prove"
```

No paid AI in Excel needed: file to a chat → a workbook with formulas back → open in Excel.

Required: one statement that sounds plausible but the data does not prove.

Templates and rubric — in L08_05_Homework and demo_v2/student_starter.

SUMMARY

What to take away

1. **The describe prompt first**
   “describe the file, suggest 5 questions, change nothing”
2. **Formulas in words**
   “Excel formulas, not pasted numbers”
3. **One chart per question**
   title = finding, one accent colour
4. **The report as a file**
   open the downloaded file, attach the recipe
5. **60 seconds of checks**
   rows, one point, percent of what

Next lesson — L09: the Claude ecosystem and artifacts. The same analysis becomes an interactive page opened by a link.